

Experimental Validation

This section demonstrates the practical viability of CIF’s formal mechanisms through empirical evaluation across production multiagent architectures. We present experimental setup (sec:exp-setup) and key findings (sec:key-findings). Detailed statistical analysis, ablation studies, and scalability metrics are provided in sec:extended-results.

Experimental Setup

Target Architectures

We evaluated CIF across six production multiagent systems representing diverse architectural patterns:

Table 1: Multiagent system architectures evaluated.

System	Architecture	Communication
Claude Code	Hierarchical ($1 + n$)	Task delegation
AutoGPT	Autonomous + plugins	Tool-based
CrewAI	Role-based (3–10)	Sequential/parallel
LangGraph	Graph-based	State machine